

Replicating Table 2, Panel B (5-Fold) from Chernozhukov et al. (2018)

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June 1, 2026

GitHub Repository: <https://github.com/manojrsan/dml-replication>

1 The Original Paper

This study replicates a specific result from *Double/Debiased Machine Learning for Treatment and Structural Parameters* by Victor Chernozhukov, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins, published in *The Econometrics Journal* in 2018 (Chernozhukov et al., 2018).

The paper addresses a fundamental problem in causal inference: when the number of controls is large relative to the sample size, naïvely plugging machine learning estimates of nuisance functions into a structural equation produces estimators that are biased at the $\sqrt{N}$ rate due to regularization bias and overfitting. The authors show that this bias can be removed by combining two ingredients: (1) *Neyman-orthogonal* moment conditions, which have reduced sensitivity to errors in nuisance estimation, and (2) *cross-fitting*, an efficient form of sample-splitting that prevents overfitting from contaminating the structural estimate. The resulting procedure is called Double/Debiased Machine Learning (DML).

2 The Result of Interest

This replication targets the 5-fold cross-fitting row of Panel B, the partially linear regression model, from Table 2 of Chernozhukov et al. (2018). This result reports the estimated Average Treatment Effect (ATE) of 401(k) eligibility (`e401`) on net total financial assets (`net_tfa`) across several machine learning methods used for nuisance estimation. The first portion of Table 1 reports the original paper’s target values, while the second portion reports my replication estimates.

This particular result is worth reproducing because it is one of the paper’s clearest empirical demonstrations of the DML procedure. It translates the method into a concrete, policy-relevant estimate using a well-known dataset and shows how the structural estimate changes, or remains stable, when different nuisance learners are used.

3 Data and Methods

Data

The data come from the 1991 Survey of Income and Program Participation (SIPP), following the sample construction of Abadie (2003). The raw file `input/sipp1991.dta` is loaded by `code/preprocess.R`, which selects the eleven analysis variables and drops incomplete cases before saving the analysis-ready dataset to `temp/clean_data.rds`. The outcome variable is net total financial assets (`net_tfa`) and the treatment indicator is whether the household’s employer offers a 401(k) plan (`e401`). The covariates are age, income, education, family size, marital status, two-earner status, defined benefit pension status, IRA participation, and home ownership.

Methods

The partially linear regression (PLR) model is:

$$Y = D\theta_0 + g_0(X) + U, \quad E[U \mid X, D] = 0, \quad (1)$$

$$D = m_0(X) + V, \quad E[V \mid X] = 0, \quad (2)$$

where Y is net financial assets, D is 401(k) eligibility, X is the vector of controls, and θ_0 is the ATE of interest. The nuisance functions g_0 and m_0 are estimated using cross-fitting. The analysis uses Lasso, Regression Tree, Random Forest, Boosting, Neural Network, and an Ensemble average for nuisance estimation. For Lasso, I use a rich polynomial and interaction feature matrix designed to follow the original replication code as closely as possible. I use the PLR orthogonal score with the DML2 cross-fitting procedure throughout. Results are aggregated across repeated random sample splits, reporting the median ATE and two standard error measures.

4 Replication Results

Table 1 presents the replicated results alongside the paper’s targets. In plain terms, the estimates imply that 401(k) eligibility has a positive and economically meaningful effect on net financial assets. The fact that the estimated effect is broadly stable across several different machine learning methods illustrates the main empirical point of the DML approach: once the nuisance functions are estimated flexibly and the orthogonal score is used, the final structural estimate is less dependent on any single nuisance specification.

Table 1: Replication of Table 2, Panel B (5-Fold): Estimated ATE of 401(k) Eligibility on Net Financial Assets (Partially Linear DML).

	Lasso	Reg. Tree	Forest	Boosting	Neural Net	Ensemble	Best
<i>Panel A: Paper targets (Chernozhukov et al., 2018)</i>							
ATE	8187	8871	9247	9110	9038	9166	9215
	[1558]	[1418]	[1328]	[1328]	[1355]	[1310]	[1312]
	(1298)	(1358)	(1295)	(1314)	(1322)	(1299)	(1294)
<i>Panel B: This replication</i>							
ATE	9676	9026	9269	8983	9213	9338	9667
	[1365]	[1394]	[1328]	[1331]	[1649]	[1330]	[1366]
	(1347)	(1356)	(1308)	(1325)	(1495)	(1311)	(1345)

The replication is closest for the tree-based and ensemble methods, while the largest discrepancy appears for Lasso. This pattern is consistent with the fact that the Lasso specification is especially sensitive to the exact polynomial and interaction feature construction used in the original code. The neural network estimate is also somewhat sensitive to random initialization and repeated sample splitting. Overall, however, the replication follows the same general pattern as the original table: the ATE is positive for every learner, the estimates are of similar economic magnitude, and the two standard error rows move closely with the corresponding rows in the published result. This suggests that the score function, cross-fitting, and aggregation steps are implemented consistently with the original replication code.

5 What I Learned

The most straightforward aspect of the replication was the DML estimator itself. Once the cross-fitting loop and orthogonal score are correctly coded, the structural estimates can be computed in a systematic way and the aggregation across repeated splits is mechanical. The framework’s robustness to learner choice, its central theoretical claim, is also directly visible

in the results: once nuisance functions are estimated reasonably well, θ_0 is stable across very different machine learning methods.

The hardest part was matching the nuisance function specification. The paper’s original R code contains a compact one-line formula (`x1`) for the Lasso feature matrix that expands to a large set of polynomial terms and interactions, but this detail is much more visible in the code than in the paper text. Without reading `401K.R` carefully, it would be easy to run Lasso on the raw covariates and obtain a noticeably different estimate, incorrectly concluding that the replication failed. Computational cost was a secondary challenge because repeated cross-fitting requires many machine learning fits. Exact seed matching with the original is also difficult because the original code uses parallel execution with per-worker seeds. These issues point to the same lesson: code-level documentation matters as much as methodological description. If I were the original author, I would explicitly state the feature construction for penalised learners and the parallelisation setup in a replication appendix.

References

- Abadie, A. (2003). Semiparametric instrumental variable estimation of treatment response models. *Journal of Econometrics*, 113(2):231–263.
- Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., and Robins, J. (2018). Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68.